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HEAVY DUTY TIRE

Abstract

A heavy duty tire includes a tread, sidewalls, beads, a carcass, a belt, chafers, and a tag holder including an RFID tag. In the tire, the carcass includes a carcass ply including carcass cords, and the carcass cords are formed from an organic fiber. The carcass ply includes a ply body extending between the beads and turned-up portions each connected to the ply body and turned up around the bead. In a standard ground-contact state, an end of the turned-up portion is axially inward of an axially outer end of a rim, and the RFID tag is axially outward of the axially outer end of the rim and is radially outward of a radially outer end of the rim. In the tire, an angle formed between a length direction of the RFID tag and the carcass cord is not less than 80 degrees and not greater than 90 degrees.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION

[0001] The present application claims priority to Japanese Application No. JP 2024-033667, filed on Mar. 6, 2024, the entire contents of which are incorporated herein by reference.

FIELD

[0002] The present disclosure relates to a heavy duty tire.

BACKGROUND ART

[0003] In order to manage data regarding manufacturing management, customer information, running history, etc., of tires, tires on which radio frequency identification (RFID) tags that perform reading and writing via radio waves are mounted, have been considered. Various studies have been conducted for a technology to mount an RFID tag to a tire (for example, Japanese Laid-Open Patent Publication No. 2021-046057).

[0004] Japanese Laid-Open Patent Publication No. 2021-046057 proposes a technology to mount an RFID tag to a heavy duty tire used for a truck or a bus.

[0005] Meanwhile, steel cords are generally adopted as tire cords, such as carcass cords, in heavy duty tires used for trucks and buses.

[0006] In the case where an RFID tag is mounted to a tire in which steel cords are adopted as carcass cords, the steel cords may become a barrier to radio waves, so that reading of data from the RFID tag could be easily impeded.

[0007] One or more embodiments may provide a heavy duty tire having good performance in reading data from an RFID tag and good durability.

[0008] In the present disclosure, good durability of the tire means that the tire has a long runnable distance and the RFID tag is less likely to be damaged.

SUMMARY

[0009] A heavy duty tire according to one aspect of the present disclosure includes: [0010] a tread; [0011] a pair of sidewalls each connected to an end of the tread and located radially inward of the tread; [0012] a pair of beads each located radially inward of the sidewall; [0013] a carcass located inward of the tread and the pair of sidewalls and extending on and between one bead and the other bead; [0014] a belt stacked on the carcass on a radially inner side of the tread; [0015] a pair of chafers each located radially inward of the sidewall and configured to come into contact with a rim; and [0016] a tag member including an RFID tag, wherein [0017] the bead includes a core and an apex located radially outward of the core, [0018] the carcass includes a carcass ply including a plurality of carcass cords aligned with each other, [0019] the carcass cords are formed from an organic fiber, [0020] the carcass ply includes a ply body extending between the pair of beads and a pair of turned-up portions each connected to the ply body and turned up around the bead, [0021] the rim is a standardized rim, [0022] a state where the tire is fitted on the rim and an internal pressure of the tire is adjusted to a standardized internal pressure is a standardized state, [0023] a state where a load that is 50% of the standardized load is applied to the tire in the standardized state and the tire is brought into contact with a flat surface is a standard ground-contact state, [0024] in the standard ground-contact state, an end of the turned-up portion is located inward of an axially outer end of the rim in an axial direction, [0025] in the standard ground-contact state, the RFID tag is located outward of the axially outer end of the rim in the axial direction and is located outward of a radially outer end of the rim in a radial direction, and [0026] an angle formed between a length direction of the RFID tag and the carcass cord is not less than 80 degrees and not greater than 90

degrees.
[0027] According to the present disclosure, it is possible to provide a heavy duty tire having good performance in reading data from an RFID tag and good durability.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
[0028] FIG. **1** is a cross-sectional view showing a part of a heavy duty tire according to one embodiment of the present disclosure;
[0029] FIG. **2** is a cross-sectional view showing a part of the heavy duty tire in FIG. **1**;
[0030] FIG. **3** is a plan view of a tag member;
[0031] FIG. **4** is a cross-sectional view taken along a line IV-IV in FIG. **3**;
[0032] FIG. **5** is a cross-sectional view showing a part of the heavy duty tire in a ground-contact state;
[0033] FIG. **6** illustrates an angle formed between the length direction of an RFID tag and a carcass cord; and
[0034] FIG. **7** is a cross-sectional view showing a part of a heavy duty tire according to another embodiment of the present disclosure.
DETAILED DESCRIPTION
[0035] The tire of the present disclosure may be fitted on a rim. The interior of the tire may be filled with air to adjust the internal pressure of the tire. The tire fitted on the rim is also referred to as tire-rim assembly. The tire-rim assembly may include the rim and the tire fitted on the rim.
[0036] In the present disclosure, a state where a tire is fitted on a standardized rim, the internal pressure of the tire is adjusted to a standardized internal pressure, and no load is applied to the tire may be referred to as standardized state.
[0037] In the present disclosure, unless otherwise specified, the dimensions and angles of each component of the tire are measured in the standardized state.
[0038] The dimensions and angles of each component in a meridian cross-section of the tire, which cannot be measured in a state where the tire is fitted on the standardized rim, are measured in a cut plane of the tire obtained by cutting the tire along a plane including a rotation axis. In this measurement, the tire may be set such that the distance between right and left beads is equal to the distance between the beads in the tire that is fitted on the standardized rim. The configuration of the tire that cannot be confirmed in a state where the tire is fitted on the standardized rim may be confirmed in the above-described cut plane.
[0039] The standardized rim means a rim specified in a standard on which the tire is based. The “standard rim” in the JATMA standard, the “Design Rim” in the TRA standard, and the “Measuring Rim” in the ETRTO standard are standardized rims.
[0040] The standardized internal pressure means an internal pressure specified in the standard on which the tire is based. The “highest air pressure” in the JATMA standard, the “maximum value” recited in the “TIRE LOAD LIMITS AT VARIOUS COLD INFLATION PRESSURES” in the TRA standard, and the “INFLATION PRESSURE” in the ETRTO standard are standardized internal pressures.
[0041] A standardized load means a load specified in the standard on which the tire is based. The “maximum load capacity” in the JATMA standard, the “maximum value” recited in the “TIRE LOAD LIMITS AT VARIOUS COLD INFLATION PRESSURES” in the TRA standard, and the “LOAD CAPACITY” in the ETRTO standard are standardized loads.
[0042] In the present disclosure, a crosslinked rubber may be a molded product obtained by pressurizing and heating a rubber composition. The crosslinked rubber may be a crosslinked product of the rubber composition. The rubber composition may be a material obtained by mixing a

base rubber and chemicals in a kneading machine such as a Banbury mixer. The base rubber of the rubber composition may not be crosslinked. The base rubber of the crosslinked rubber may be crosslinked. The crosslinked rubber may also be referred to as vulcanized rubber, and the rubber composition may also be referred to as unvulcanized rubber.

[0043] Examples of the base rubber may include natural rubber (NR), butadiene rubber (BR), styrene-butadiene rubber (SBR), isoprene rubber (IR), ethylene-propylene rubber (EPDM), chloroprene rubber (CR), acrylonitrile-butadiene rubber (NBR), and isobutylene-isoprene-rubber (IIR). Examples of the chemicals may include reinforcing agents such as carbon black and silica, plasticizers such as aromatic oil, fillers such as zinc oxide, lubricants such as stearic acid, antioxidants, processing aids, sulfur, and vulcanization accelerators. Selection of a base rubber and chemicals, the amounts of the selected chemicals, etc., may be determined as appropriate according to the specifications of components, such as a tread and a sidewall, for which the rubber composition is used.

[0044] In the present disclosure, a tire may include a tread portion, a pair of bead portions, and a pair of sidewall portions as portions thereof. The tread portion may be a portion of the tire that comes into contact with a road surface. Each bead portion may be a portion of the tire that is fitted to a rim. Each sidewall portion may be a portion of the tire that extends between the tread portion and the bead portion.

[0045] In the present disclosure, a complex elastic modulus of a component formed from a crosslinked rubber, of the components included in the tire, may be measured according to the standards of JIS K6394. The measurement conditions are as follows. [0046] Initial strain=10% [0047] Dynamic strain=±1% [0048] Frequency=10 Hz [0049] Mode=stretch mode [0050] Temperature=70° C.

[0051] In this measurement, a test piece (a length of 40 mm× a width of 4 mm× a thickness of 1 mm) is sampled from the tire. The length direction of the test piece is caused to coincide with the circumferential direction of the tire. When a test piece cannot be sampled from the tire, a test piece is sampled from a sheet-shaped crosslinked rubber (hereinafter, also referred to as rubber sheet) obtained by pressurizing and heating a rubber composition, which is used for forming the component to be measured, at a temperature of 170° C. for 12 minutes.

[0052] In the present disclosure, the complex elastic modulus is represented as a complex elastic modulus at 70° C.

[Findings on which Present Disclosure is Based]

[0053] In the case where an RFID tag is mounted to a heavy duty tire, the RFID tag may be generally embedded in a bead portion.

[0054] Heavy duty tires used for trucks and buses may be all-steel tires in which steel cords are used as a reinforcing material for both a carcass and a belt due to the conditions of use. The amount of steel (metal) in an all-steel tire may be large, so that the exchange of radio waves by an RFID tag mounted on the all-steel tire could be impeded by the metal and the all-steel tire could have poor reading performance.

[0055] In the case where a heavy duty tire on which an RFID tag is mounted on a vehicle, depending on the mounting position of the RFID tag on the tire, the exchange of radio waves could be impeded by a metal rim such as an aluminum wheel, and as a result, the reading operation could become difficult.

[0056] Regarding addressing these issues, for the latter, it is possible to consider mounting the RFID tag at a position away from the metal wheel, which can address the issue.

[0057] Meanwhile, for the former, it is possible to consider replacing cords made of steel (hereinafter referred to as steel cords) with cords made of an organic fiber (hereinafter referred to as organic fiber cords). In the case where organic fiber cords are used as carcass cords instead of steel cords, the strain in the vicinity of an end of each turned-up portion of a carcass could become larger, thereby reducing the durability of a heavy duty tire.

[0058] Therefore, the present inventors have examined ensuring good reading performance of an RFID tag without impairing the durability of a heavy duty tire.
[0059] As a result, the present inventors have found that it is possible to ensure good reading performance of an RFID tag without impairing the durability of a heavy duty tire, by mounting a tag holder or tag member including the RFID tag to the tire at a predetermined position in a predetermined orientation, adopting organic fiber cords as carcass cords, and providing a carcass such that an end of a turned-up portion of a carcass ply is at a predetermined position, and thus have completed the disclosure described below.
DETAILS OF EMBODIMENTS OF PRESENT DISCLOSURE
[0060] Hereinafter, the present disclosure will be described in detail based on embodiments with appropriate reference to the drawings.
[0061] FIG. **1** shows a part of a heavy duty tire **2** (hereinafter also referred to simply as “tire **2**”), according to one embodiment of the present disclosure. The tire **2** may be mounted to a vehicle such as a truck or a bus.
[0062] In FIG. **1**, the tire **2** may be fitted on a rim R (standardized rim).
[0063] FIG. **1** shows a part of a cross-section (hereinafter referred to as meridian cross-section) of the tire **2** along a plane including the rotation axis of the tire **2**. In FIG. **1**, the right-left direction is the axial direction of the tire **2**, and the up-down direction is the radial direction of the tire **2**. The direction perpendicular to the surface of the drawing sheet of FIG. **1** is or includes the circumferential direction of the tire **2**.
[0064] FIG. **2** shows a part of the cross-section shown in FIG. **1**.
[0065] In FIG. **1**, an alternate long and short dash line CL extending in the radial direction represents the equator plane of the tire **2**. In FIG. **1** and FIG. **2**, a solid line BBL extending in the axial direction is a bead base line. The bead base line BBL is a line that defines the rim diameter (see JATMA or the like) of the rim R.
[0066] The tire **2** may include a tread **4**, a pair of sidewalls **6**, a pair of chafers **8**, a pair of beads **10**, a carcass **12**, a pair of cushion layers **14**, a pair of steel fillers **16**, an inner liner **18**, a reinforcing layer **20**, and a tag holder or tag member **26**.
[0067] The tread **4** may be located radially outward of the carcass **12**. The tread **4** may come into contact with a road surface at a tread surface **24** thereof. At least three circumferential grooves **28** may be formed on the tread **4**. Accordingly, at least four land portions **30** may be formed in the tread **4**.
[0068] On the tread **4** of the tire **2** shown in FIG. **1**, three circumferential grooves **28** may be formed, whereby four land portions **30** may be formed therein. These land portions **30** may be aligned in the axial direction and extend continuously in the circumferential direction.
[0069] Among the three circumferential grooves **28** formed on the tread **4**, the circumferential groove **28** located on each outer side in the axial direction may be a shoulder circumferential groove **28***s*. The circumferential groove **28** sandwiched between two shoulder circumferential grooves **28***s* in the axial direction and located on the equator plane CL may be a center circumferential groove **28***c*. In the tire **2**, the three circumferential grooves **28** may be composed of such a center circumferential groove **28***c* and a pair of such shoulder circumferential grooves **28***s*.
[0070] Among the four land portions **30** formed in the tread **4**, the land portion **30** located on each outer side in the axial direction may be a shoulder land portion **30***s*. The shoulder land portion **30***s* may be located outward of the shoulder circumferential groove **28***s* in the axial direction and may include an end PE of the tread surface **24**. The land portion **30** located inward of the shoulder land portion **30***s* in the axial direction may be a middle land portion **30***m*. The middle land portion **30***m* may be located between the center circumferential groove **28***c* and the shoulder circumferential groove **28***s* in the axial direction. In the tire **2**, the four land portions **30** may be composed of a pair of such middle land portions **30***m* and a pair of such shoulder land portions **30***s*.
[0071] The tread **4** may include a base portion **32** and a cap portion **34** located radially outward of

the base portion **32**. The base portion **32** may be formed from a crosslinked rubber that has low heat generation properties. The cap portion **34** may be formed from a crosslinked rubber for which wear resistance and grip performance are taken into consideration. As shown in FIG. **1**, the base portion **32** may cover the entire reinforcing layer **20**. The cap portion **34** may cover the entire base portion **32**. The cap portion **34** may include the tread surface **24**.
[0072] In FIG. **1**, a position indicated by reference character PC corresponds to an equator. The equator PC is the point of intersection of the tread surface **24** and the equator plane CL. In the case where the groove **28** is located on the equator plane CL as in the tire **2**, the equator PC is specified on the basis of a virtual tread surface obtained on the assumption that the groove **28** is not present thereon.
[0073] The distance in the radial direction, from the bead base line BBL to the equator PC, obtained in the tire **2** in the standardized state is the cross-sectional height (see JATMA or the like) of the tire **2**.
[0074] Each sidewall **6** may be connected to an end of the tread **4**. The sidewall **6** may be located radially inward of the tread **4**. The sidewall **6** may be located axially outward of the carcass **12**. A position indicated by reference character PS is an inner end of the sidewall **6**.
[0075] The sidewall **6** may be formed from a crosslinked rubber for which cut resistance is taken into consideration. The complex elastic modulus of the sidewall **6** may be not less than 2.0 MPa and not greater than 6.0 MPa.
[0076] A position indicated by reference character PW is an axially outer end of the tire **2** (hereinafter referred to as outer end PW). In the case where decorations such as patterns and letters are present on the outer surface of the tire **2**, the outer end PW may be specified on the basis of a virtual outer surface obtained on the assumption that the decorations are not present thereon. The tire **2** may have a maximum width at the outer end PW. The outer end PW may also be referred to as maximum width position.
[0077] In the present disclosure, the maximum width position PW in the tire **2** in the standardized state is a reference maximum width position PWb. The distance in the axial direction from one reference maximum width position PWb to another reference maximum width position PWb is the cross-sectional width (see JATMA or the like) of the tire **2**.
[0078] In FIG. **1**, a length indicated by reference character H is the distance in the radial direction from the bead base line BBL to the reference maximum width position PWb. The distance H in the radial direction is also referred to as radial height of the reference maximum width position PWb.
[0079] In the tire **2** in the standardized state, the ratio of the radial height H of the reference maximum width position PWb to the cross-sectional height may be not less than 0.40 and not greater than 0.60.
[0080] Each chafer **8** may be located radially inward of the sidewall **6**. The chafer **8** may come into contact with the rim R. A position indicated by reference character PB is an outer end of the chafer **8**.
[0081] The chafer **8** may be formed from a crosslinked rubber for which wear resistance is taken into consideration. The complex elastic modulus of the chafer **8** may be not less than 10 MPa and not greater than 15 MPa. The chafer **8** may be harder than the sidewall **6**.
[0082] Each bead **10** may be located axially inward of the chafer **8**. The bead **10** may be located radially inward of the sidewall **6**. The bead **10** may include a core **36** and an apex **38**.
[0083] The core **36** may extend in the circumferential direction. The core **36** may include a core body **36***m* and a wrapping layer **36***r*.
[0084] The core body **36***m* may be a ring extending in the circumferential direction. The core body **36***m* may include a wire made of steel and wound in the circumferential direction. A cross-sectional shape of the core body **36***m* may be shaped by winding the wire in a regular manner. Accordingly, in a cross-section of the core body **36***m*, cross-sectional units each including a plurality of wire cross-sections aligned substantially in the axial direction are stacked in a plurality of stages

substantially in the radial direction. The cross-sectional shape of the core body **36***m* may be represented by a line that circumscribes the core body **36***m*. In an implementation, as shown in FIG. **1**, the core body **36***m* may have a hexagonal cross-sectional shape. In an implementation, the core body **36***m* may have a quadrangular cross-sectional shape.

[0085] The core body **36***m* may generally have six side surfaces **36***ms*. As shown in FIG. **1**, one side surface **36**msb out of the six side surfaces **36***ms* may be located so as to face a seat Rs of the rim R. In the present embodiment, the side surface **36**msb which is located so as to face the seat Rs of the rim R may be a bottom surface of the core body **36***m*. The core body **36***m* may have the bottom surface **36**msb which is located so as to face the seat Rs of the rim R. In the meridian cross-section of the tire **2**, the contour of the bottom surface **36**msb may be represented by a straight line.

[0086] The wrapping layer **36***r* may surround the core body **36***m*. The wrapping layer **36***r* may cover the core body **36***m*. The wrapping layer **36***r* may help prevent the core body **36***m* from falling apart.

[0087] The configuration of the wrapping layer **36***r* may be a suitable configuration in which the wrapping layer **36***r* may help prevent the core body **36***m* from falling apart. The wrapping layer **36***r* may be composed of a cord helically wound around the core body **36***m*, a rubberized fabric wrapped around the core body **36***m*, or the like.

[0088] The apex **38** may be located radially outward of the core **36**. The apex **38** may extend radially outward from the core **36**. The apex **38** may be tapered outward. An outer end PA of the apex **38** may be located radially outward of the outer end PB of the chafer **8**.

[0089] The apex **38** may include an inner apex **40** and an outer apex **42**. The inner apex **40** may be located radially outward of the core **36**. The outer apex **42** may be located radially outward of the inner apex **40**.

[0090] The inner apex **40** may be tapered outward. The inner apex **40** may be formed from a hard crosslinked rubber. The complex elastic modulus of the inner apex **40** may be not less than 60 MPa and not greater than 90 MPa.

[0091] The outer apex **42** may be thick around an outer end PU of the inner apex **40**. The outer apex **42** may be tapered inward and tapered outward from the thick portion thereof.

[0092] An inner end PG**1** of the outer apex **42** may be located near the core **36**. An outer end PG**2** of the outer apex **42** may also be the outer end PA of the apex **38**.

[0093] The outer apex **42** may be formed from a crosslinked rubber. The outer apex **42** may be softer than the inner apex **40**. The complex elastic modulus of the outer apex **42** may be not less than 3.0 MPa and not greater than 6.0 MPa.

[0094] The carcass **12** may be located inward of the tread **4**, the pair of sidewalls **6**, and the pair of chafers **8**. The carcass **12** may extend on and between the pair of beads **10**. The carcass **12** of the tire **2** may have a radial structure.

[0095] The carcass **12** may include at least one carcass ply **44**. The carcass **12** of the tire **2** may be composed of one carcass ply **44**. The carcass ply **44** may be turned up around each bead **10**.

[0096] The carcass ply **44** may include a ply body **48** and a pair of turned-up portions **50**. The ply body **48** may extend between the pair of beads **10**, e.g., between one bead **10** and the other bead **10**. Each turned-up portion **50** may be connected to the ply body **48** and turned up around the bead **10**. The turned-up portion **50** of the tire **2** may be turned up around the bead **10** from the inner side toward the outer side in the axial direction. The turned-up portion **50** may be turned up such that an end PF of the turned-up portion **50** is located axially outward of the apex **38**.

[0097] In the tire **2** according to the present embodiment, the end PF of the turned-up portion **50** may be at a predetermined position. The specific position will be described below.

[0098] In the tire **2**, each bead **10** may be between the ply body **48** and the turned-up portion **50**.

[0099] The carcass ply **44** may include a large number of carcass cords aligned with each other, which are not shown in FIGS. **1** and **2**. These carcass cords may be covered with a topping rubber. Each carcass cord may intersect the equator plane CL. The material of the carcass cord may be an organic fiber. The carcass cord may be an organic fiber cord. Examples of the organic fiber may include nylon fibers, polyester fibers, rayon fibers, and aramid fibers.

[0100] In the tire **2**, the carcass cords may be organic fiber cords, and blocking of radio waves, exchanged by an RFID tag **54**, by the carcass cords may be suppressed.

[0101] In an implementation, as the organic fiber cords, cords made of an aramid fiber (hereinafter referred to as "aramid cords") may be used, with a view toward having durability and heat resistance similar to steel cords.

[0102] In an implementation, the aramid fiber that is the material of the aramid cords may be a para-aramid fiber. In an implementation, as the carcass cords adopted in the tire **2**, cords made of a para-aramid fiber (hereinafter referred to as "para-aramid cords") may be used.

[0103] In FIG. **1**, a length indicated by reference character N is the distance in the radial direction from the bead base line BBL to the end PF of the turned-up portion **50**. The distance N in the radial direction is also referred to as radial height of the end PF of the turned-up portion **50**.

[0104] In the tire **2**, the ratio (N/H) of the radial height N of the end PF of the turned-up portion **50** to the radial height H of the reference maximum width position PWb may be not less than 0.20 and not greater than 0.40.

[0105] Each cushion layer **14** may be located between the reinforcing layer **20** and the carcass **12** at an end of the reinforcing layer **20**. The cushion layer **14** may be formed from a soft crosslinked rubber.

[0106] Each steel filler **16** may be located at a bead portion. The steel filler **16** may be turned up from the inner side toward the outer side in the axial direction around each core **36** along the carcass ply **44**. The steel filler **16** may be placed so as to wrap a radially inner portion of the bead **10** from the radially inner side of the turned-up portion **50**.

[0107] The steel filler **16** may include a large number of filler cords aligned with each other. The material of the filler cords may be steel. The steel filler **16** may include steel cords. In the steel filler **16**, the steel cords may be covered with a topping rubber.

[0108] An outer end **16***f* of the steel filler **16** may be located axially outward of the turned-up portion **50**. The outer end **16***f* may be located radially inward of the end PF of the turned-up portion **50**. An inner end **16***s* of the steel filler **16** may be located axially inward of the ply body **48**. The position in the radial direction of the inner end **16***s* only needs to substantially coincide with that of the outer end **16***f*, and the inner end **16***s* may be located radially outward of the outer end **16***f* or may be located radially inward of the outer end **16***f*.

[0109] The inner liner **18** may be located inward of the carcass **12**. The inner liner **18** may be joined to the inner surface of the carcass **12** via an insulation formed from a crosslinked rubber. The inner liner **18** may form an inner surface of the tire **2**. The inner liner **18** may be formed from a crosslinked rubber that has an excellent air blocking property.

[0110] The reinforcing layer **20** may be located inward of the tread **4** in the radial direction. The reinforcing layer **20** may be located between the carcass **12** and the tread **4**. The reinforcing layer **20** may include a belt **62** and a band **70**. The stiffness of a tread portion may be increased by the reinforcing layer **20**.

[0111] The belt **62** may include a plurality of belt plies **64** aligned in the radial direction. Each belt ply **64** may be placed such that both ends thereof are opposed to each other across the equator plane CL. In an implementation, the belt **62** of the tire **2** may include four belt plies **64**. The four belt plies **64** may include a first belt ply **64**A located on the inner side in the radial direction, a second belt ply **64**B located outward of the first belt ply **64**A, a third belt ply **64**C located outward of the second belt ply **64**B, and a fourth belt ply **64**D located outward of the third belt ply **64**C.

[0112] In the tire **2**, the second belt ply **64**B may have a widest width in the axial direction, and the fourth belt ply **64**D may have a narrowest width in the axial direction. The first belt ply **64**A and the third belt ply **64**C may have the same width in the axial direction, or the width in the axial direction of the first belt ply **64**A may be wider than the width in the axial direction of the third belt ply **64**C.

[0113] An end **62***e* of the belt **62** of the tire **2** may be represented as an end of the belt ply **64** having a widest width in the axial direction among the plurality of belt plies **64** included in the belt **62**. In the tire **2**, as described above, among the four belt plies **64** included in the belt **62**, the second belt ply **64**B may have a widest width in the axial direction. The end **62***e* of the belt **62** of the tire **2** may be represented as an end of the second belt ply **64**B having a widest width in the axial direction.

[0114] As shown in FIG. **1**, an end of the first belt ply **64**A, the end of the second belt ply **64**B, and an end of the third belt ply **64**C may be located outward of the shoulder circumferential groove **28***s* in the axial direction. An end of the fourth belt ply **64**D may be located inward of the shoulder circumferential groove **28***s* in the axial direction.

[0115] In the tire **2**, each belt ply **64** included in the belt **62** may include a large number of belt cords aligned with each other. The belt cords may be covered with a topping rubber. The belt cords of the tire **2** may be steel cords.

[0116] In the tire **2**, the density of the belt cords in each belt ply **64** may be not less than 15 ends/5 cm and not greater than 30 ends/5 cm. The density of the belt cords may be represented as the number of cross-sections of the belt cords included per 5 cm width of the belt ply **64** in a cross-section of the belt ply **64** along a plane perpendicular to the direction in which the belt cords extend.

[0117] In each belt ply **64**, the belt cords may be inclined with respect to the circumferential direction. In the tire **2**, as the direction of inclination of the belt cords included in each belt ply **64** with respect to the circumferential direction (hereinafter referred to as inclination direction of the belt cords), any direction can be selected independently for each belt ply **64**. In the tire **2**, from the viewpoint of ensuring a stable ground-contact shape, the inclination direction of the belt cords of the second belt ply **64**B may be opposite to the inclination direction of the belt cords of the third belt ply **64**C.

[0118] The inclination angle of the belt cords included in each belt ply **64** with respect to the circumferential direction may be appropriately selected within a range of, e.g., not less than 10 degrees and not greater than 60 degrees.

[0119] The band **70** may have two ends **70***e* opposed to each other across the equator plane CL. The band **70** may include a helically wound band cord. The band cord may be covered with a topping rubber.

[0120] In the tire **2**, the band cord may be a steel cord or an organic fiber cord. Examples of the material of the organic fiber cord may include nylon fibers, polyester fibers, rayon fibers, and aramid fibers.

[0121] In an implementation, the band cord may be a steel cord.

[0122] As described above, the band **70** may include the helically wound band cord. The band **70** may have a jointless structure. In the band **70**, an angle of the band cord with respect to the circumferential direction of the tire **2** may be, e.g., not greater than 5 degrees or not greater than 2 degrees. The band cord of the band **70** may extend substantially in the circumferential direction.

[0123] The density of the band cord in the band **70** may be not less than 20 ends/5 cm and not greater than 35 ends/5 cm. The density of the band cord may be represented as the number of cross-sections of the band cord included per 5 cm width of the band **70** in a cross-section of the band **70** along a plane perpendicular to the direction in which the band cord extends.

[0124] In the tire **2**, each of the end of the second belt ply **64**B and the end of the third belt ply **64**C may be covered with a rubber layer **66**. Between the end of the second belt ply **64**B and the end of the third belt ply **64**C each covered with the rubber layer **66**, two rubber layers **66** may be further placed. In the tire **2**, an edge member **68** composed of a total of four rubber layers **66** may be formed between the end of the second belt ply **64**B and the end of the third belt ply **64**C. The edge member **68** may be formed from a crosslinked rubber. The edge member **68** may contribute to maintaining the interval between the end of the second belt ply **64**B and the end of the third belt ply

64C. In the tire **2**, a change in the positional relationship between the end of the second belt ply **64**B and the end of the third belt ply **64**C due to running may be suppressed. The edge member **68** may be a part of the reinforcing layer **20**. The reinforcing layer **20** of the tire **2** may include a pair of such edge members **68** in addition to the belt **62** and the band **70**.

[0125] As described above, the band **70** may have the two ends **70***e* opposed to each other across the equator plane CL. The band **70** may extend in the axial direction from the equator plane CL toward each end **70***e*. Each end **70***e* of the band **70** may be located outward of the shoulder circumferential groove **28***s* in the axial direction. In the radial direction, the band **70** may be located inward of each shoulder circumferential groove **28***s*.

[0126] In the tire **2**, the band **70** may help suppress growth of the carcass **12** due to running and may help suppresses a shape change of the tire **2**. Therefore, an increase in strain in the vicinity of the end PF of each turned-up portion **50** of the carcass **12** during running may be suppressed. Therefore, the durability of the tire **2** may be better than that of a tire in which no band **70** is provided.

[0127] In the tire **2**, each end **70***e* of the band **70** may be located inward of the end **62***e* of the belt **62** in the axial direction. The belt **62** may be wider than the band **70**. The belt **62** may constrain each end **70***e* of the band **70**. The belt **62** may help suppress fluctuation of the tension of the band cord included in the band **70**. Occurrence of a break of the band cord due to tension fluctuation may be suppressed, so that the band **70** may stably exhibit the function of suppressing a shape change. In an implementation, each end **70***e* of the band **70** may be located inward of the end **62***e* of the belt **62** in the axial direction.

[0128] A force may act on the band **70** of the tire **2** so as to spread from the inner side toward the outer side in the radial direction. This force may cause tension in the band cord of the band **70**.

[0129] In the tire **2**, in the radial direction, the second belt ply **64**B may be located inward of the band **70**, and the third belt ply **64**C may be located outward of the band **70**. In the tire **2**, the band **70** may be interposed between the second belt ply **64**B and the third belt ply **64**C. The second belt ply **64**B may be wider than the band **70**. The third belt ply **64**C may also be wider than the band **70**. The plurality of belt plies **64** included in the belt **62** of the tire **2** may include two belt plies **64** having a width wider than the width of the band **70**, and the band **70** may be interposed between these belt plies **64** having a wide width. In the tire **2**, fluctuation of the tension of the band cord included in the band **70** may be more effectively suppressed, so that a break may be less likely to occur in the band cord of the band **70**. The band **70** of the tire **2** can stably exhibit the function of suppressing a shape change. In an implementation, in the tire **2**, the plurality of belt plies **64** included in the belt **62** may include two belt plies **64** having a width wider than the width of the band **70**, and the band **70** may be interposed between these belt plies **64** having a wide width.

[0130] In the tire **2**, the first belt ply **64**A, the second belt ply **64**B, and the third belt ply **64**C may have a width wider than the width of the band **70**. In the radial direction, the first belt ply **64**A and the second belt ply **64**B may be located inward of the band **70**, and the third belt ply **64**C and the fourth belt ply **64**D may be located outward of the band **70**.

[0131] The tag member **26** may be located axially outward of the bead **10**. In the tire **2**, the tag member **26** may be provided only on the side of one sidewall **6**. In an implementation, the tag member **26** may be provided on each of the side of one sidewall **6** and the side of the other sidewall **6**. In an implementation, from the viewpoint of reducing the risk of damage, the tag member **26** may be provided only on the side of one sidewall **6** out of the pair of sidewalls **6**.

[0132] FIG. **3** is a plan view of the tag member **26**. FIG. **4** is a cross-sectional view taken along a line IV-IV in FIG. **3**.

[0133] The tag member **26** may have a plate shape. The tag member **26** may be long in a length direction thereof and short in a width direction thereof. As shown in FIG. **1**, in the tire **2**, the tag member **26** may be placed such that a first end **26***s* in the length direction thereof is located on or facing the radially outer side in the tire **2** and a second end **26***u* in the length direction thereof is

located on or facing the radially inner side in the tire **2**. In the tire **2**, the first end **26***s* may also be referred to as outer end, and the second end **26***u* may also be referred to as inner end.
[0134] The tag member **26** may include the RFID tag **54**. In FIG. **3**, for convenience of description, the RFID tag **54** is shown by a solid line, and the entirety thereof may be covered with a protector **56**. The tag member **26** may include the RFID tag **54** and the protector **56**. The RFID tag **54** may be located at the center of the tag member **26**. The protector **56** may be formed from a crosslinked rubber. The protector **56** may have stiffness substantially equal to the stiffness of the outer apex **42**. In the tire **2**, formation of a good communication environment may be considered, and a crosslinked rubber having high electrical resistance may be used for the protector **56**. The protector **56** may be formed from a rubber that has high insulation properties.
[0135] In an implementation, the RFID tag **54** may be a small and lightweight electronic component that includes, e.g., a semiconductor chip **58** obtained by making a transmitter/receiver circuit, a control circuit, a memory, or the like, into a chip; and an antenna **60**. Upon receiving interrogation radio waves, the RFID tag **54** may use the radio waves as electrical energy and transmits various data in the memory as response radio waves. The RFID tag **54** may be a type of passive radio frequency identification transponder.
[0136] In the RFID tag **54**, the antenna **60** may extend in the length direction of the RFID tag **54** (the right-left direction in FIG. **3**) from the semiconductor chip **58**. As shown in FIG. **3**, the RFID tag **54** may include a pair of such antennas **60** provided so as to extend in the length direction thereof, and the semiconductor chip **58** located between the pair of antennas **60**. In the tag member **26**, the RFID tag **54** may be placed such that the length direction of the RFID tag **54** coincides with the length direction of the tag member **26**.
[0137] The shape of each antenna **60** shown in FIG. **3** is an example, and each antenna **60** included in the RFID tag **54** can take various shapes. The shape of each antenna **60** may be, e.g., a shape extending while bending, a helical shape, or the like.
[0138] In the embodiment of the present disclosure, the external dimensions (length, width, and thickness) of the RFID tag **54** are represented as the length, width, and thickness of the circumscribed rectangular parallelepiped of the RFID tag **54**. The length of the RFID tag **54** may be equal to or larger than the width of the RFID tag **54**. The width of the RFID tag **54** may be equal to or larger than the thickness of the RFID tag **54**.
[0139] In addition, in the embodiment of the present disclosure, the length direction of the RFID tag **54** may also be the length direction of the antennas **60** included in the RFID tag **54**. In an implementation, the length direction of the antennas **60** is the length direction of the RFID tag **54**.
[0140] The tag member **26** may be a plate-shaped member in which the RFID tag **54** is covered with a crosslinked rubber (protector). In an implementation, from the viewpoint of reducing the risk of damage to the RFID tag **54** and forming a good communication environment, the thickness of the tag member **26** in the tire **2** may be not less than 1.0 mm and not greater than 2.5 mm. The thickness of the tag member **26** in the tire **2** may be represented as the maximum thickness of the tag member **26** at the semiconductor chip **58** of the RFID tag **54**.
[0141] A length TL of the tag member **26** before embedding in the tire **2** may be not less than 60 mm and not greater than 80 mm, and a width TW thereof may be not less than 10 mm and not greater than 20 mm.
[0142] FIG. **5** shows a part of a meridian cross-section of the tire **2** in a ground-contact state. The cross-section in FIG. **5** is a trace of a cross-sectional image of the tire **2** taken, e.g., by a computer tomography method using X-rays (hereinafter referred to as X-ray CT method) in a state where a load that is 50% of the standardized load is applied to the tire **2** in the standardized state and the tire **2** is brought into contact with a flat surface FS.
[0143] In the present disclosure, the state where a load that is 50% of the standardized load is applied to the tire **2** in the standardized state and the tire **2** is brought into contact with the flat surface FS is also referred to as standard ground-contact state.

[0144] In the standard ground-contact state shown in FIG. **5**, the camber angle of the tire **2** is set to 0 degrees.
[0145] In FIG. **5**, a position indicated by reference character PRa is an axially outer end of a flange Rf of the rim R. An alternate long and short dash line indicated by reference character LRa is a straight line passing through the axially outer end PRa and extending in the radial direction.
[0146] A position indicated by reference character PRr is a radially outer end of the flange Rf. An alternate long and short dash line indicated by reference character LRr is a straight line passing through the radially outer end PRr and extending in the axial direction.
[0147] In FIG. **5**, a position indicated by reference character PWs is the maximum width position of the tire **2** in the standard ground-contact state. An alternate long and short dash line indicated by reference character LWsa is a straight line passing through the maximum width position PWs and extending in the radial direction. An alternate long and short dash line indicated by reference character LWsr is a straight line passing through the maximum width position PWs and extending in the axial direction.
[0148] In FIG. **5**, the tire **2** is shown in a simplified manner, and some of the components of the tire **2** shown in FIG. **1**, such as the reinforcing layer **20**, may be omitted.
[0149] As shown in FIG. **5**, in the tire **2** in the standard ground-contact state, the RFID tag **54** may be located outward of the radially outer end PRr of the rim R in the radial direction. Therefore, radio waves may not be blocked by the rim R, and the performance in reading data from the RFID tag **54** may be good. In an implementation, in the standard ground-contact state, the RFID tag **54** may be located inward of the maximum width position PWs of the tire **2** in the radial direction. In this case, the RFID tag **54** may be placed at a position where bending is less likely to occur during running, so that the RFID tag **54** itself may be less likely to be damaged, and damage to the tire **2** starting from the location where the RFID tag **54** is placed may also be less likely to occur.
[0150] In the present disclosure, the fact that the RFID tag **54** is located outward of the radially outer end PRr of the rim R may mean that the entire semiconductor chip **58** of the RFID tag **54** is located outward of the radially outer end PRr of the rim R. In addition, the fact that the RFID tag **54** is located inward of the maximum width position PWs of the tire **2** may mean that the entire semiconductor chip **58** of the RFID tag **54** is located inward of the maximum width position PWs of the tire **2**.
[0151] Moreover, in the tire **2** in the standard ground-contact state, in the axial direction, the RFID tag **54** may be located outward of the axially outer end PRa of the rim R, and the end PF of the turned-up portion **50** may be located inward of the axially outer end PRa of the rim R. In this case, the RFID tag **54** and the end PF of the turned-up portion **50** may be located away from each other in the axial direction. Therefore, an increase in strain in the vicinity of the end PF of the turned-up portion **50**, which could otherwise occur due to the RFID tag **54**, foreign matter for the tire, being close to the end PF, can be suppressed.
[0152] In the present disclosure, the fact that the RFID tag **54** is located outward of the axially outer end PRa of the rim R means that the entire semiconductor chip **58** of the RFID tag **54** may be located outward of the axially outer end PRa of the rim R.
[0153] In an implementation, from the above-described viewpoint, the mounting position of the tag member **26** mounted on the tire **2** may be a position at which the entire semiconductor chip **58** of the RFID tag **54** is included in a region surrounded by the alternate long and short dash lines LWsr and LWsa and the alternate long and short dash lines LRr and LRa in the standard ground-contact state.
[0154] Furthermore, in the tire **2**, the RFID tag **54** may be placed away from the vicinity of the end of the flange Rf and the vicinity of the maximum width position PWs where large strain could occur if the RFID tag **54** were to be present therein. In the tire **2**, promotion of an increase in strain by the presence of the RFID tag **54** may be suppressed.
[0155] In the tire **2**, occurrence of damage due to the presence of the RFID tag **54** may be

suppressed. Moreover, damage to the RFID tag **54** itself may also be suppressed.

[0156] In the tire **2**, the RFID tag **54** may be incorporated therein, and good durability may be maintained.

[0157] In FIG. **5**, a double-headed arrow LA indicates the distance in the axial direction between the end PF of the turned-up portion **50** and the axially outer end PRa of the rim R. A double-headed arrow LB indicates the distance in the axial direction between the RFID tag **54** and the axially outer end PRa of the rim R. Here, the distance LB in the axial direction between the RFID tag **54** and the axially outer end PRa of the rim R may be the shortest distance in the axial direction between the semiconductor chip **58** of the RFID tag **54** and the axially outer end PRa of the rim R.

[0158] In the tire **2**, the distance LA in the axial direction and the distance LB in the axial direction may satisfy LA<LB. As already described, in order to ensure the reading performance of the RFID tag **54** while suppressing an increase in strain in the vicinity of the end PF of the turned-up portion **50**, the distance in the axial direction between the end PF of the turned-up portion **50** and the RFID tag **54** may be large, but from the viewpoint of more easily ensuring the reading performance of the RFID tag **54**, the RFID tag **54** may be far away from the rim R in the axial direction.

[0159] From the above-described viewpoint, the distance LB in the axial direction may be not less than 1.5 times and not greater than 8.0 times the distance LA in the axial direction.

[0160] The tire **2** may be a tire in which, in the standard ground-contact state shown in FIG. **5**, the end PE of the tread surface **24** is located axially outward of the alternate long and short dash line indicated by reference character LRa.

[0161] The tire **2** may be a tire in which, in the standard ground-contact state shown in FIG. **5**, the outer end PU of the inner apex **40** is located axially inward of the alternate long and short dash line indicated by reference character LRa.

[0162] FIG. **6** illustrates an angle formed between the length direction of the RFID tag **54** and the carcass cord. FIG. **6** illustrates the positional relationship between the RFID tag **54** and the carcass cords when a part of the tire **2** is viewed in the axial direction. Therefore, in FIG. **6**, the direction penetrating the drawing sheet thereof is the axial direction of the tire **2**. As the angle formed between the length direction of the RFID tag **54** and the carcass cord, an angle on the acute side is adopted, except when this angle is 90 degrees.

[0163] In the tire **2**, the RFID tag **54** may be mounted in a predetermined orientation. In an implementation, an angle θ formed between a carcass cord **46** (included in the carcass **12**) and the length direction of the RFID tag **54** may be not less than 80 degrees and not greater than 90 degrees.

[0164] In the tire **2**, an organic fiber cord may be adopted as each carcass cord **46**. In a tire in which an organic fiber cord is used as each carcass cord **46**, each sidewall portion of the tire may bend more easily than in a tire in which a steel cord is used as each carcass cord. Therefore, in the case where the RFID tag **54** is mounted to the tire in which an organic fiber cord is used as each carcass cord, depending on the orientation of the RFID tag **54**, the RFID tag **54** may be likely to be significantly influenced by bending of the sidewall portion and could be easily damaged.

[0165] Under such circumstances, the RFID tag **54** may be placed such that the angle formed between the length direction thereof and the carcass cord **46** is close to 90 degrees, and the above-described influence of bending on the RFID tag **54** can be reduced.

[0166] In an implementation, in the tire **2**, the angle θ formed between the length direction of the RFID tag **54** and the carcass cord **46** may be set to be not less than 80 degrees and not greater than 90 degrees.

[0167] In an implementation, the angle θ may be closer to 90 degrees.

[0168] Here, the angle θ means the angle formed between the length direction of the RFID tag **54** and a carcass cord **46**A closest to the center of the semiconductor chip **58** of the RFID tag **54** when the tire **2** is viewed in the axial direction.

[0169] The center of the semiconductor chip **58** when the tire **2** is viewed in the axial direction

means the center of the circumscribed rectangle of the semiconductor chip **58** (the point of intersection of the diagonals of the circumscribed rectangle) when viewed in the axial direction.
[0170] In FIG. **1**, a length indicated by a double-headed arrow t is the shortest distance from the outer surface of the tire **2** to the RFID tag **54**. The shortest distance t may be the thickness of the rubber covering the RFID tag **54**.
[0171] In the tire **2**, the shortest distance t may be not less than 3.5 mm. Accordingly, the RFID tag **54** may be covered with rubber having a sufficient thickness. In the tire **2**, promotion of an increase in strain by the presence of the RFID tag **54** may be effectively suppressed. In the tire **2**, occurrence of damage due to the presence of the RFID tag **54** may be effectively suppressed. In an implementation, the shortest distance t may be not less than 4.0 mm. The upper limit of the shortest distance t may depend on the position of the RFID tag **54**, and the upper limit of the shortest distance t may be a suitable upper limit.
[0172] In the tire **2**, the tag member **26** may be located axially outward of the outer apex **42** on the radially outer side of the end PF of the turned-up portion **50**. In this case, the tag member **26** may be in contact with the outer apex **42**. The boundary between the tag member **26** and the outer apex **42** may form a part of the outer surface of the outer apex **42**. In an implementation, the boundary between the tag member **26** and the outer apex **42** may form a part of the outer surface of the apex **38**.
[0173] In the tire **2**, the RFID tag **54** may be located between the outer end PG**2** of the outer apex **42** and the end PF of the turned-up portion **50** in the radial direction.
[0174] The RFID tag **54** of the tire **2** may be placed in the bead portion where the degree of bending is small. In the tire **2**, the risk of damage to the RFID tag **54** may be low.
[0175] In the tire **2**, the outer apex **42**, which may be softer than the inner apex **40**, may be located axially inward of the RFID tag **54**, and promotion of an increase in strain by the presence of the RFID tag **54** may be effectively suppressed. In the tire **2**, occurrence of damage due to the presence of the RFID tag **54** may be suppressed. Moreover, damage to the RFID tag **54** itself may also be suppressed. In the tire **2**, the RFID tag **54** may be incorporated therein, and good durability may be maintained.
[0176] In an implementation, from the viewpoint of maintaining good durability while achieving formation of a good communication environment and reduction of the risk of damage to the RFID tag **54**, the tag member **26** may be in contact with the outer apex **42** on the radially outer side of the end PF of the turned-up portion **50**, and the RFID tag **54** may be located between the outer end PG**2** of the outer apex **42** and the end PF of the turned-up portion **50** in the radial direction. In an implementation, from the same viewpoint, the RFID tag **54** may be located between the outer end PG**2** of the outer apex **42** and the outer end PB of the chafer **8** in the radial direction.
[0177] In the tire **2**, the entire tag member **26** may be located radially outward of the outer end PB of the chafer **8**. The interference of the tag member **26** with the outer end PB of the chafer **8** may be effectively suppressed, so that waving (i.e., creasing) of the outer end PB of the chafer **8** may be effectively suppressed. Promotion of an increase in strain by the presence of the RFID tag **54** may be effectively suppressed. In the tire **2**, occurrence of damage due to the presence of the RFID tag **54** may be suppressed. Moreover, damage to the RFID tag **54** itself may also be suppressed. In the tire **2**, the RFID tag **54** may be incorporated therein, and good durability may be maintained.
[0178] In the tire **2**, the entire tag member **26** may be located radially inward of the outer end PG**2** of the outer apex **42**. Accordingly, the influence of the tag member **26** on bending of the sidewall portion may be effectively suppressed. In the tire **2**, good durability and ride comfort may be maintained. In an implementation, from this viewpoint, the outer end **26***s* of the tag member **26** may be located radially inward of the outer end PG**2** of the outer apex **42**.
[0179] In the tire **2**, the outer end PU of the inner apex **40** may be located between the end PF of the turned-up portion **50** and the RFID tag **54** in the radial direction. Accordingly, the hard inner apex **40** may help effectively increase the stiffness of the bead portion. The strain acting on the

RFID tag **54** may be effectively reduced. Promotion of an increase in strain by the presence of the RFID tag **54** may be effectively suppressed. In an implementation, from this viewpoint, the outer end PU of the inner apex **40** may be located between the end PF of the turned-up portion **50** and the RFID tag **54** in the radial direction. In this case, from the viewpoint of being able to more effectively suppress promotion of an increase in strain by the presence of the RFID tag **54**, the outer end PU of the inner apex **40** may be located between the end PF of the turned-up portion **50** and the RFID tag **54** in the radial direction, and further the outer end PU of the inner apex **40** may be located between the end PF of the turned-up portion **50** and the outer end PB of the chafer **8** in the radial direction.

[0180] In an implementation, as described above, the tire **2** may be a heavy duty tire in which, in each bead portion, the end PF of the turned-up portion **50** may be located axially outward of the apex **38**, or the structure of each bead portion in the heavy duty tire may have another suitable structure.

[0181] FIG. **7** is a cross-sectional view showing a part of a heavy duty tire **102** (hereinafter referred to as tire **102**) according to another embodiment of the present disclosure.

[0182] The tire **102** may have the same configuration as the tire **2**, except that the position of the end of the turned-up portion of the carcass ply in each bead portion may be different. In FIG. **7**, components other than a carcass ply **144** and a bead **110** are designated by the same reference characters as in FIG. **2**.

[0183] The bead portion of the tire **102** shown in FIG. **7** may have a structure in which: a turned-up portion **150** of the carcass ply **144** is wound around a core **136** substantially in a single turn while being turned up from the inner side toward the outer side in the axial direction on the radially inner side of the bead **110**; and an end RF**2** of the turned-up portion **150** along a radially upper surface **136***j* of the core **136** may be between the core **136** and an apex **138** (sometimes referred to as a bead-winding structure). In the tire **102**, the apex **138** may have an inner apex **140** and an outer apex **142**, and the end RF**2** of the turned-up portion **150** may be between the radially upper surface **136***j* of the core **136** (wrapping layer **136***r*) and the inner apex **140**.

[0184] In FIG. **7**, reference character **136***m* denotes a core body, and reference character **148** denotes a ply body.

[0185] In the tire **102**, the end RF**2** of the turned-up portion **150** may be between the core **136** and the inner apex **140** which may be harder than the outer apex **142** and the chafer **8**. Therefore, in the tire **102**, strain in the vicinity of the end RF**2** of the turned-up portion **150** may be particularly less likely to occur.

[0186] In the tire **102**, an organic fiber cord, which is more flexible than a steel cord, may be adopted as each carcass cord. Therefore, in the tire **102**, strong bending back of each turned-up portion (so-called spring back) in a green tire molding process or the like, which could be likely to occur in the case if a steel cord were to be adopted as each carcass cord in a tire having a bead-winding structure, may be less likely to occur.

[0187] In the tire **102** in which an organic fiber cord is adopted as each carcass cord, the above-described spring back may be less likely to occur, and molding defects such as formation of cavities in bead portions in a green tire molding process may be less likely to occur.

[0188] In the tire **102**, the tag member **26** having the RFID tag **54** may be mounted at a predetermined position as in the tire **2**. In an implementation, the tire **102** may also be a heavy duty tire having good performance in reading data from an RFID tag and having good durability.

[0189] In the tire **102**, the core **136** may be further coated with a vulcanized rubber layer around the wrapping layer **136***r*.

[0190] According to the present disclosure, the heavy duty tire **2**, which has good performance in reading data from an RFID tag and has good durability, may be obtained.

[0191] The above-described technology can be applied when mounting RFID tags to various tires.

Additional Note

[0192] The present disclosure includes aspects described below.
[0193] [1] A heavy duty tire including: [0194] a tread; [0195] a pair of sidewalls each connected to an end of the tread and located radially inward of the tread; [0196] a pair of beads each located radially inward of the sidewall; [0197] a carcass located inward of the tread and the pair of sidewalls and extending on and between one bead and the other bead; [0198] a belt stacked on the carcass on a radially inner side of the tread; [0199] a pair of chafers each located radially inward of the sidewall and configured to come into contact with a rim; and [0200] a tag member including an RFID tag, wherein [0201] the bead includes a core and an apex located radially outward of the core, [0202] the carcass includes a carcass ply including a plurality of carcass cords aligned with each other, [0203] the carcass cords are formed from an organic fiber, [0204] the carcass ply includes a ply body extending between the pair of beads and a pair of turned-up portions each connected to the ply body and turned up around the bead, [0205] the rim is a standardized rim, [0206] a state where the tire is fitted on the rim and an internal pressure of the tire is adjusted to a standardized internal pressure is a standardized state, [0207] a state where a load that is 50% of the standardized load is applied to the tire in the standardized state and the tire is brought into contact with a flat surface is a standard ground-contact state, [0208] in the standard ground-contact state, an end of the turned-up portion is located inward of an axially outer end of the rim in an axial direction, [0209] in the standard ground-contact state, the RFID tag is located outward of the axially outer end of the rim in the axial direction and is located outward of a radially outer end of the rim in a radial direction, and [0210] an angle formed between a length direction of the RFID tag and the carcass cord is not less than 80 degrees and not greater than 90 degrees.
[0211] [2] The heavy duty tire according to [**1**] above, wherein the carcass cords are formed from an aramid fiber.
[0212] [3] The heavy duty tire according to [1] or [**2**] above, wherein the RFID tag is located inward of a maximum width position of the tire in the radial direction in the standard ground-contact state.
[0213] [4] The heavy duty tire according to any one of [1] to [**3**] above, further comprising a band located between the tread and the belt in the radial direction and having two ends opposed to each other across an equator plane.
[0214] [5] The heavy duty tire according to any one of [1] to [**4**] above, wherein the end of the turned-up portion is interposed between the core and the apex.
[0215] [6] The heavy duty tire according to any one of [1] to [**4**] above, wherein [0216] the end of the turned-up portion is located axially outward of the apex, and [0217] a distance LB in the axial direction between the RFID tag and the axially outer end of the rim in the standard ground-contact state is longer than a distance LA in the axial direction between the end of the turned-up portion and the axially outer end of the rim in the standard ground-contact state.

Claims

1. A heavy duty tire, comprising: a tread; a pair of sidewalls each connected to an end of the tread and located radially inward of the tread; a pair of beads each located radially inward of the sidewall; a carcass located inward of the tread and the pair of sidewalls and extending on and between one bead and another bead; a belt stacked on the carcass on a radially inner side of the tread; a pair of chafers each located radially inward of the sidewall and configured to come into contact with a rim; and a tag holder including an RFID tag, wherein the beads each include a core and an apex located radially outward of the core, the carcass includes a carcass ply including a plurality of carcass cords aligned with each other, the carcass cords are formed from an organic fiber, the carcass ply includes a ply body extending between the pair of beads and a pair of turned-up portions each connected to the ply body and turned up around the bead, the rim is a standardized rim, a state where the tire is fitted on the rim and an internal pressure of the tire is adjusted to a standardized internal pressure is a standardized state, a state where a load that is 50% of a standardized load is applied to the tire in the standardized state and the tire is brought into contact with a flat surface is a standard ground-contact state, in the standard ground-contact state, ends of the turned-up portions are located inward of an axially outer end of the rim in an axial direction, in the standard ground-contact state, the RFID tag is located outward of the axially outer end of the rim in the axial direction and is located outward of a radially outer end of the rim in a radial direction, and an angle formed between a length direction of the RFID tag and a length direction of the carcass cord is not less than 80 degrees and not greater than 90 degrees.

2. The heavy duty tire according to claim 1, wherein the carcass cords are formed from an aramid fiber.

3. The heavy duty tire according to claim 1, wherein the RFID tag is located inward of a maximum width position of the tire in the radial direction in the standard ground-contact state.

4. The heavy duty tire according to claim 1, further comprising a band located between the tread and the belt in the radial direction and having two ends opposed to each other across an equator plane.

5. The heavy duty tire according to claim 1, wherein the ends of the turned-up portions are between the core and the apex.

6. The heavy duty tire according to claim 1, wherein: the ends of the turned-up portions are located axially outward of the apex, and a distance LB in the axial direction between the RFID tag and the axially outer end of the rim in the standard ground-contact state is longer than a distance LA in the axial direction between the end of the turned-up portion and the axially outer end of the rim in the standard ground-contact state.

7. The heavy duty tire according to claim 1, wherein, in the standard ground-contact state, the RFID tag is located outward of the radially outer end of the rim in the radial direction and is located inward of a maximum width position of the heavy duty tire in the radial direction.

8. The heavy duty tire according to claim 1, wherein an angle formed between a length direction of the RFID tag and a length direction of the carcass cord is 90 degrees.

9. The heavy duty tire according to claim 1, wherein a distance in the axial direction between the ends of the turned-up portions and an axially outer end of the rim is less than a distance in the axial direction between the RFID tag and the axially outer end of the rim.

10. The heavy duty tire according to claim 9, wherein the distance in the axial direction between the RFID tag and the axially outer end of the rim is not less than 1.5 times and not greater than 8.0 times the distance in the axial direction between the ends of the turned-up portions and the axially outer end of the rim.

11. A heavy duty tire that is mountable on a standardized rim such that a state where a load that is 50% of a standardized load is applied to the heavy duty tire in a state where the heavy duty tire is fitted on the standardized rim and an internal pressure of the tire is adjusted to a standardized internal pressure, and the tire is brought into contact with a flat surface is a standard ground-contact state, the heavy duty tire comprising: a tread; a pair of sidewalls each connected to an end of the tread and located radially inward of the tread; a pair of beads each located radially inward of the sidewall; a carcass located inward of the tread and the pair of sidewalls and extending on and between one bead and another bead; a belt stacked on the carcass on a radially inner side of the tread; a pair of chafers each located radially inward of the sidewall and configured to come into contact with a rim; and a tag holder including an RFID tag, wherein the beads each include a core and an apex located radially outward of the core, the carcass includes a carcass ply including a plurality of carcass cords aligned with each other, the carcass cords are formed from an organic fiber, the carcass ply includes a ply body extending between the pair of beads and a pair of turned-up portions each connected to the ply body and turned up around the beads, in the standard ground-contact state, ends of the turned-up portions are located inward of an axially outer end of the rim in an axial direction, in the standard ground-contact state, the RFID tag is located outward of the axially outer end of the rim in the axial direction and is located outward of a radially outer end of the rim in a radial direction, and an angle formed between a length direction of the RFID tag and a length direction of the carcass cord is not less than 80 degrees and not greater than 90 degrees.

12. The heavy duty tire according to claim 11, wherein the carcass cords are formed from an aramid fiber.

13. The heavy duty tire according to claim 11, wherein the RFID tag is located inward of a maximum width position of the tire in the radial direction in the standard ground-contact state.

14. The heavy duty tire according to claim 11, further comprising a band located between the tread and the belt in the radial direction and having two ends opposed to each other across an equator plane.

15. The heavy duty tire according to claim 11, wherein the ends of the turned-up portions are between the core and the apex.

16. The heavy duty tire according to claim 11, wherein: the ends of the turned-up portions are located axially outward of the apex, and a distance LB in the axial direction between the RFID tag and the axially outer end of the rim in the standard ground-contact state is longer than a distance LA in the axial direction between the end of the turned-up portion and the axially outer end of the rim in the standard ground-contact state.

17. The heavy duty tire according to claim 11, wherein, in the standard ground-contact state, the RFID tag is located outward of the radially outer end of the rim in the radial direction and is located inward of a maximum width position of the heavy duty tire in the radial direction.

18. The heavy duty tire according to claim 11, wherein an angle formed between a length direction of the RFID tag and a length direction of the carcass cord is 90 degrees.

19. The heavy duty tire according to claim 11, wherein a distance in the axial direction between the ends of the turned-up portions and an axially outer end of the rim is less than a distance in the axial direction between the RFID tag and the axially outer end of the rim.

20. The heavy duty tire according to claim 19, wherein the distance in the axial direction between the RFID tag and the axially outer end of the rim is not less than 1.5 times and not greater than 8.0 times the distance in the axial direction between the ends of the turned-up portions and the axially outer end of the rim.